

Authentication & OTP Login



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Passwordless phone-based identity for the Ethiopian market

Category: Identity & Access | Platform: Flutter 3 / Laravel 12 | Status: Production

OTP + Sanctum Auth method	0 stored Passwords	Android relay SMS channel	EN / AM / TI Languages
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1. What it is

A frictionless, passwordless login system that authenticates users by their phone number and a one-time code. WaveMart deliberately avoids email/password flows because smartphone ownership in Ethiopia is near universal while email adoption is low, and OTP dramatically reduces the cost of fraud and account recovery.

2. How it works

01 Send OTP 1	User enters phone -> POST /api/auth/send-otp -> backend creates OtpCode row and triggers SMS relay.
02 Deliver SMS 2	An Android device running the WaveMart OTP service receives an HTTP poll, forwards the SMS via the local SIM.
03 Verify & login 3	User enters code -> POST /api/auth/login -> on success, Sanctum returns a personal access token stored in flutter_secure_storage.
04 Resume session 4	On relaunch, the app reads the secure token, calls /api/user, and silently restores the session.

3. Capabilities

Capability	What it does
Phone-first login	Send, resend, verify and login flows with destination phone echoed back for UI hints.
Registration flow	Two-step: capture name + gender + phone, then verify OTP. Email is optional.
Auto token refresh	Auth token persists in secure storage; expired tokens trigger a forced re-login.

Logout cleanup	Server-side token revocation, secure-storage wipe, and FCM token deletion in one call.
Multi-language UI	Login, registration, error and resend messages are translated in EN / AM / TI.
Throttled attempts	Backend enforces per-phone rate limits and OTP expiry windows.

4. Investor relevance

WHY IT MATTERS

OTP-first auth removes the single biggest drop-off point for new users in Ethiopia: forgotten passwords. Combined with the SMS relay, WaveMart achieves enterprise-grade delivery at a fraction of international gateway costs, while keeping users inside the app from first launch.

5. Tech surface

Component	Location
AuthService	lib/data/services/auth_service.dart
Login screen	lib/presentation/screens/auth/otp_login_screen.dart
Registration screen	lib/presentation/screens/auth/registration_screen.dart
Auth API controller	WaveMart/app/Http/Controllers/Api/AuthController.php
OTP service	WaveMart/app/Services/OtpService.php
Token storage	flutter_secure_storage (AES)

6. API endpoints

```
POST /api/auth/send-otp { phone_number }
POST /api/auth/resend-otp { phone_number }
POST /api/auth/login { phone_number, otp_code }
POST /api/auth/verify-otp { phone_number, otp }
POST /api/auth/register { first_name, last_name, phone_number, gender, otp_code? }
POST /api/auth/logout (Sanctum) -> revokes token + clears device FCM
```